Retail Sales Performance Analysis

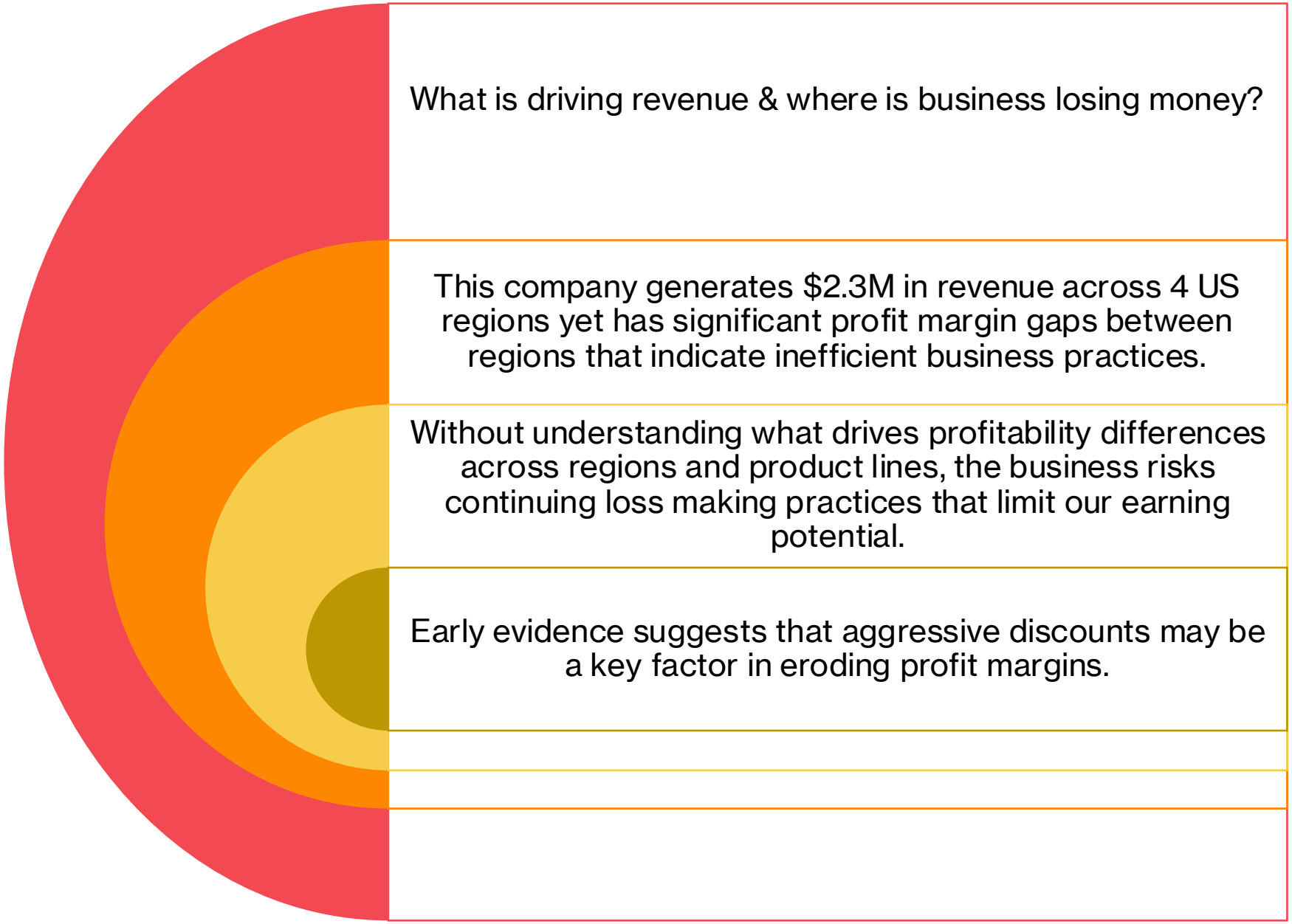
Michael Covelli | March 2026

An exploratory analysis of sales, profit and discount impact on US stores (2014 – 2017)

Agenda

- Business Problem
- Dataset Overview
- Cleaning and Validation
- Key Findings
- Recommendations
- Next Steps

Business Problem



Dataset Overview

Source: Superstore Sales (Kaggle)

10,134 rows, 23 columns

Date range: January 2014 – December 2017

Tools used: MariaDB via phpMyAdmin, Tableau Public

Column	Type	Description
Order ID	VARCHAR	Unique order identifier
cleaned_order_date	DATE	Date order was placed
Region	VARCHAR	US sales region
Category	VARCHAR	Product category
Sub-Category	VARCHAR	Product sub-category
Sales	DECIMAL	Revenue from sales
Discount	DECIMAL	Discount applied (0.0 – 0.8)
Profit	DECIMAL	Profit or loss from sales

Data Cleaning

Data Verification Checks

- Order ID -> Customer ID integrity check
  - Only one Customer ID per Order ID
- Invalid ship dates
  - All shipments occur after order date
- Whitespace validation
  - No columns with trailing or leading whitespace
- Negative sales values
  - No sales with negative values
- Null Value Check
  - No null values in critical columns
- 5/5 verification checks passed

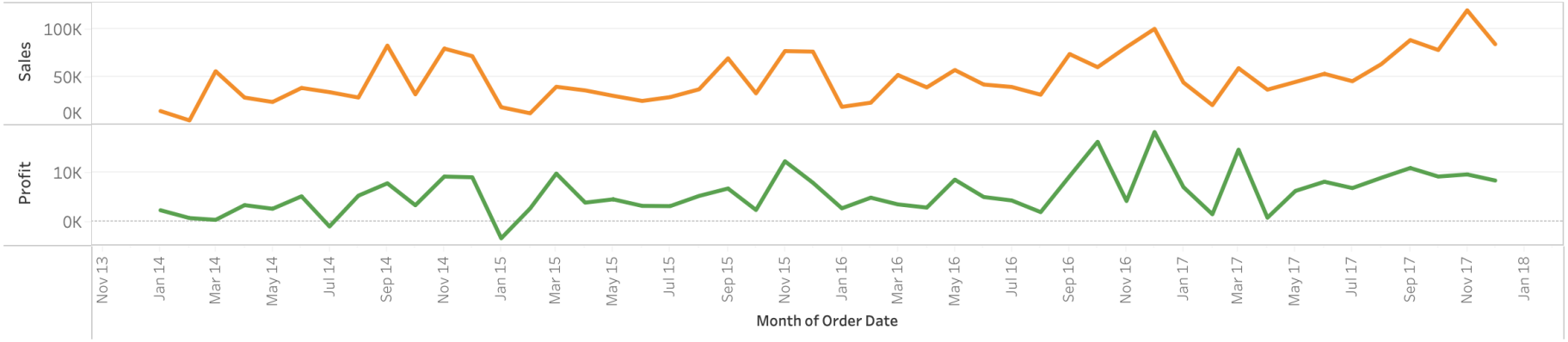
Data Type Corrections

Column	Original Type	Corrected Type	Reason
Order Date	VARCHAR(10)	DATE	Required for ship date vs order date validation
Ship Date	VARCHAR(10)	DATE	Required for time-based analysis and date comparison
Sales	DECIMAL(8,4)	DECIMAL(10,2)	Original Precision too small for large order values

Overall Performance 2014 - 2017

Total Sales	Total Profits	Profit Margin %
$2,323,628.57	$286,408.16	12.33%

Monthly Trends

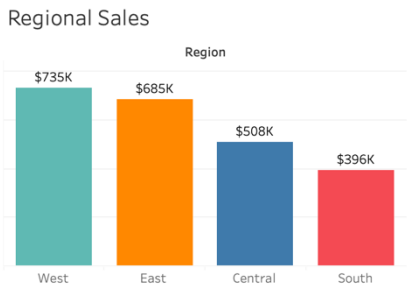


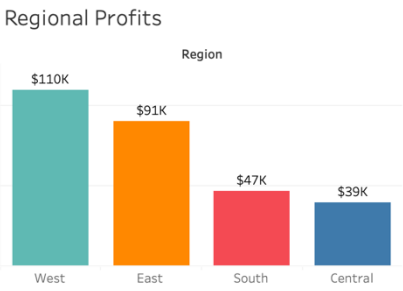
- $2.3M total revenue
- $286K total profit
- 12.33% overall margin

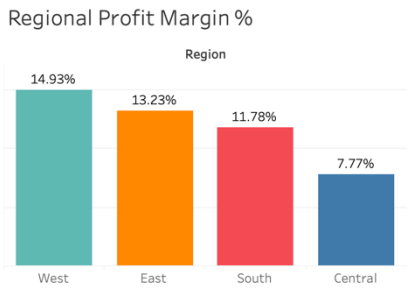
- Upward trajectory from 2014 to 2017
- Consistent Q4 spikes indicating strong seasonal demand

Regional Performance

- West leads in both Sales ($735K) and Profit Margin (14.93%)
- Central has the third highest sales ($508K) but the lowest profit margin (7.77%)
- Central has $508K in revenue but retains half compared to West

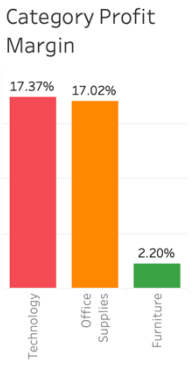


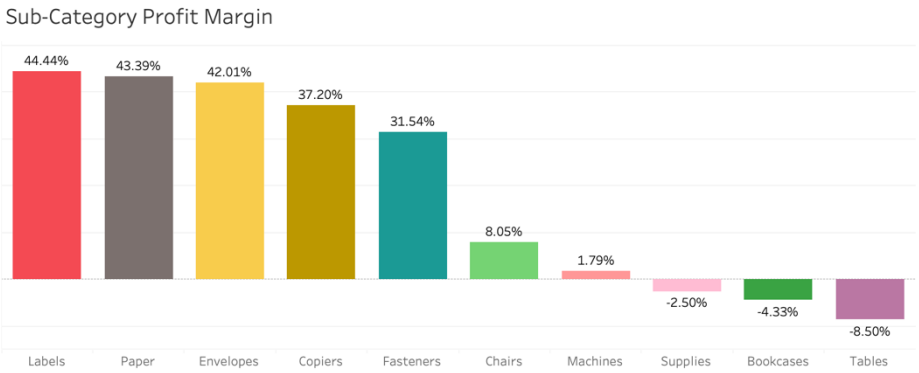




Category & Sub-Category Profitability

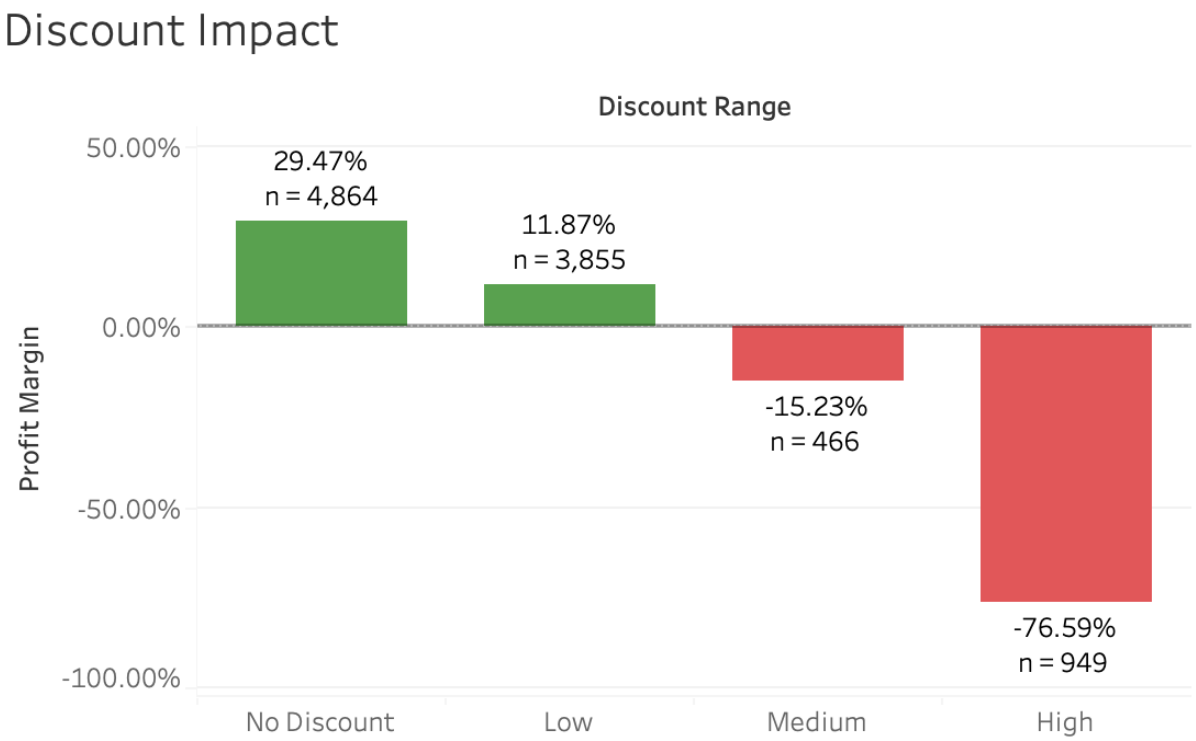
- Technology and Office Supplies have a healthy 17%+ profit margin
- Furniture is barely profitable at 2.20%
- Bookcases and Tables are loss making sub-categories dragging down furniture profits





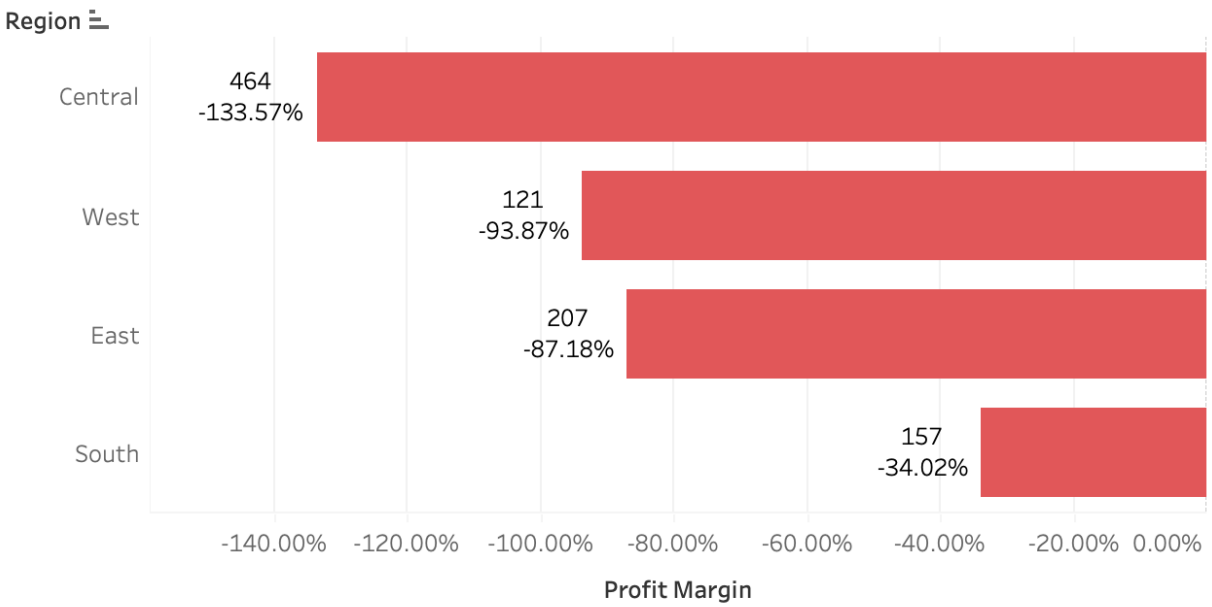
Discount Impact

- No discounts produce 29.47% profits.
- Medium discounts create profit loss.
- High discount items have catastrophically low profit margin at -76.59% across 949 orders.



Central Region Deep Dive

- 464 High Discount Orders in Central.
- 207 unique customers affected.
- -133.57% profit margin at High Discounts in Central



Recommendations

Cap discounts at 20% company wide

Medium and High discounts are universally loss making.

Audit Central region pricing practices

Highest volume of damaging discount orders.

Review furniture sub-category strategy

Tables and Bookcases are consistently losing money.

Leverage Q4 Seasonal demand

Consistent November/December spikes suggest opportunity for targeted inventory planning.

Next Steps

Are high discount customers retained long term or one time buyers?

Which sales representatives are approving the largest discounts?

Is furniture profitability salvageable with pricing adjustments or should these items be discontinued?

Expand analysis to include customer lifetime value.

Thank you

Michael Covelli

michaelrcovelli@gmail.com

github.com/mcovelli

linkedin.com/in/mcovelli

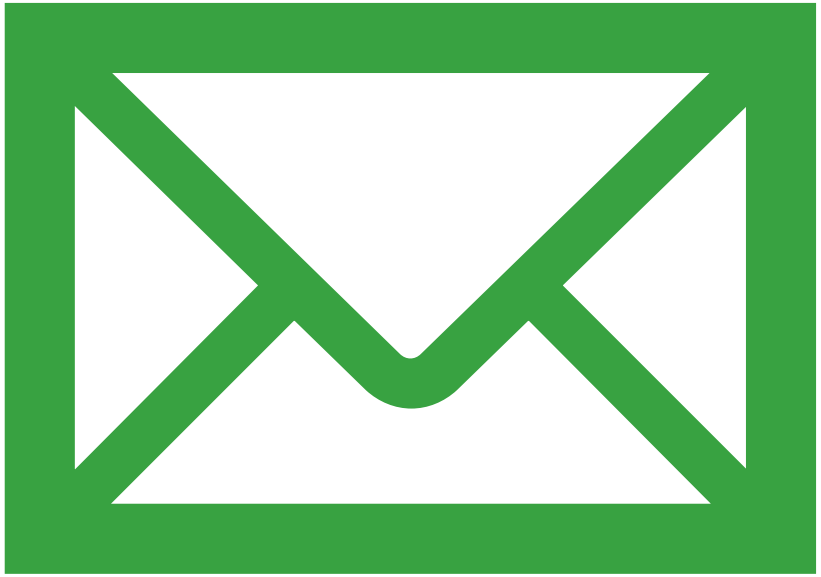



Tableau Public